1. PRODUCT AND COMPANY IDENTIFICATION

Product name : DL-alpha-Tocopheryl Acetate
Substance name : 3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-ol acetate
CAS-No. : 7695-91-2

Manufacturer or supplier's details

Company : DSM Nutritional Products Vietnam Limited

Address : No. 9 Street 14, VSIP 11-A,
Tan Uyen District, Binh Duong Province 75400
Telephone : +848422742221301
Emergency telephone number : +41 848 00 11 77 (Carechem 24 International)

Recommended use of the chemical and restrictions on use

Recommended use : For the fortification of foods
Ingredient for capsules and/or tablets
Ingredient for pharmaceutical products
Ingredient for personal care products

2. HAZARDS IDENTIFICATION

GHS Classification

Not a hazardous substance or mixture.

GHS label elements

Not a hazardous substance or mixture.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Synonyms : all-rac-α-Tocopheryl Acetate

Brief description of the product : Substance
Molecular formula : C31 H52 O3

Hazardous components

Chemical name	CAS-No.	Concentration (% w/w)
3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate	7695-91-2	>= 90 - <= 100

4. FIRST AID MEASURES

General advice : No hazards which require special first aid measures.

If inhaled : Move to fresh air.
If symptoms persist, call a physician.

In case of skin contact	:	Take off contaminated clothing and shoes immediately. Wash off with soap and plenty of water.
In case of eye contact	:	Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing.
If swallowed	:	Rinse mouth with water. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person.
Most important symptoms and effects, both acute and delayed	:	No specific symptoms known.
Notes to physician	:	Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Alcohol-resistant foam Dry chemical Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	:	High volume water jet
Specific hazards during firefighting	:	None known.
Specific extinguishing methods	:	Standard procedure for chemical fires.
Special protective equipment for firefighters	:	In the event of fire, wear self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Use personal protective equipment.
Environmental precautions	:	Try to prevent the material from entering drains or water courses.
Methods and materials for containment and cleaning up	:	Wipe up with absorbent material (e.g. cloth, fleece). Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Take necessary action to avoid static electricity discharge. Product will burn under fire conditions.
Advice on safe handling	:	For personal protection see section 8.
Conditions for safe storage	:	To maintain product quality, do not store in heat or direct sunlight. Keep container tightly closed and dry.
Materials to avoid	:	No special restrictions on storage with other products.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate	7695-91-2	TWA	8 mg/m3	DSM Internal Limit

Personal protective equipment

Respiratory protection	:	Breathing apparatus needed only when aerosol or mist is formed. No personal respiratory protective equipment normally required.
Hand protection	:	Glove material: for example nitrile rubber
Eye protection	:	Safety glasses
Skin and body protection	:	Lightweight protective clothing
Hygiene measures	:	General industrial hygiene practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance	:	viscous liquid
Colour	:	colourless - yellow
Odour	:	odourless
Odour Threshold	:	No information available.
pH	:	No data available
Melting point/range	:	-27.5 °C
Boiling point/boiling range	:	267 °C (3.2 hPa)

		184 °C (0.01 hPa)
		> 300 °C (1,013 hPa) with decomposition
Flash point	:	243 - 266 °C (DIN 51758)
Evaporation rate	:	not determined
Lower explosion limit	:	not determined
Upper explosion limit	:	not determined
Vapour pressure	:	< 0.001 hPa (20 °C)
		ca. 1.4 hPa (240 °C)
Relative vapour density	:	not determined
Density	:	ca. 0.98 g/cm3 (20 °C)
Water solubility	:	< 0.8 mg/l (20 °C; OECD Test Guideline 105) practically insoluble
Solubility in other solvents	:	Alcohol: soluble
		Chloroform: easily soluble
		Diethylether: easily soluble
		Acetone: easily soluble
Partition coefficient: n-octanol/water	:	log Pow 12.2 (calculated (citation from literature))
Auto-ignition temperature	:	not pyrophoric
Ignition temperature	:	320 °C (985 - 1,004 hPa, DIN 51794)
Thermal decomposition	:	Not relevant
Viscosity, dynamic	:	ca. 6,589 mPa.s (20 °C)
Viscosity, kinematic	:	5076 mm2/s (20 °C; OECD Test Guideline 114)
		701 mm2/s (40 °C; OECD Test Guideline 114)
Explosive properties	:	Not explosive
Oxidizing properties	:	No data available

Other information

Refractive index	:	1.494 - 1.499 (589 nm, 20 °C)
Molecular weight	:	472.73 g/mol

10. STABILITY AND REACTIVITY

Reactivity	:	No hazards to be specially mentioned.
Chemical stability	:	Stable under recommended storage conditions.
Possibility of hazardous reactions	:	Possible incompatibility with materials listed under section 10.5.
Conditions to avoid	:	Heat
Incompatible materials	:	Bases

Strong acids
Oxidizing agents

Iron salts
Silver salts
Powdered metal salts

Hazardous decomposition products : No decomposition if used as directed.

11. TOXICOLOGICAL INFORMATION

Acute oral toxicity : LD50 (Rat): > 10,000 mg/kg (OECD Test Guideline 401)
: LD50 (Mouse): > 4,000 mg/kg (OECD Test Guideline 401)

Acute dermal toxicity : LD50 (Rat): > 3,000 mg/kg

Skin irritation : No skin irritation (Rabbit, OECD Test Guideline 404)

: no phototoxic skin reaction (Guinea pig)

Eye irritation : No eye irritation (Rabbit, OECD Test Guideline 405)

Sensitisation : Does not cause skin sensitisation. (human)

: no photoallergenic skin reaction (Guinea pig)

Genotoxicity in vitro : negative (Ames test, OECD Test Guideline 471)

: not genotoxic (In vitro cytogenicity study in mammalian cells, OECD Test Guideline 473)

Genotoxicity in vivo : not genotoxic (In vivo micronucleus test, Mouse, OECD Test Guideline 474)

Carcinogenicity : (Rat , OECD Test Guideline 453)
Animal testing did not show any carcinogenic effects.

Reproductive toxicity : No toxicity to reproduction (Rat, OECD Test Guideline 415)

Teratogenicity : not teratogenic
NOAEL: > 1,600 mg/kg bw/d (Rat, OECD Test Guideline 414)
: not teratogenic
NOAEL: > 1,600 mg/kg bw/d (Rabbit, OECD Test Guideline 414)

STOT - single exposure (Acute exposure)	:	The substance or mixture is not classified as specific target organ toxicant, single exposure.
STOT - repeated exposure	:	NOAEL (Oral, Rat) : 2000 mg/kg bw/d Subacute toxicity study (28 days) (OECD Test Guideline 407)
	:	NOAEL (Oral, Rat) : 500 mg/kg bw/d Sub-chronic toxicity study (90-day) (OECD Test Guideline 408)
Further information	:	The product passes into and partly through the skin of pigs.
Aspiration toxicity	:	No aspiration toxicity classification

12. ECOLOGICAL INFORMATION

Toxicity

Toxicity to fish	:	Oncorhynchus mykiss (rainbow trout) LC50 (96 h) > 100 mg/l (nominal concentration) (OECD Test Guideline 203)
Toxicity to daphnia and other aquatic invertebrates	:	Daphnia magna (Water flea) EC50 (48 h) > 100 mg/l (nominal concentration) (OECD Test Guideline 202)
Toxicity to algae	:	Pseudokirchneriella subcapitata (microalgae) EC50 (72 h) > 100 mg/l (nominal concentration) (OECD Test Guideline 201)
Toxicity to bacteria	:	Pseudomonas putida EC10 (0.5 h) > 10,000 mg/l (nominal concentration)
	:	activated sludge EC20 (0.5 h) > 927 mg/l
Toxicity to fish (Chronic toxicity)	:	Oncorhynchus mykiss (rainbow trout) NOEC (28 d) 100 mg/l (nominal concentration) (OECD Test Guideline 215)

Persistence and degradability

Biodegradability	:	Not readily biodegradable. 17 % (28 d) (OECD Test Guideline 301F)

Bioaccumulative potential

Partition coefficient: n-octanol/water	:	log Pow 12.2 (calculated (citation from literature))

Mobility in soil

Distribution among environmental compartments	:	Adsorption, Soil log Koc 7.14 (calculated value)

Results of PBT and vPvB assessment

Assessment	:	The substance does not fulfill the PBT criteria.
	:	The substance does not fulfill the vPvB criteria.

Other adverse effects

Additional ecological information	:	There is no data available for this product.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	Offer surplus and non-recyclable solutions to a licensed disposal company.
Contaminated packaging	:	Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG
Not regulated as a dangerous good

IATA-DGR
Not regulated as a dangerous good

IMDG-Code
Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Workplace Safety and Health Act and Workplace Safety and Health (General Provisions) Regulations: This product is subjected to the SDS, labelling, PEL and other requirements in the Act/Regulations.

Fire Safety (Petroleum and Flammable Materials) Regulations	:	Not applicable

16. OTHER INFORMATION

Date format : dd.mm.yyyy

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System
ACGIH = American Conference of Governmental Industrial Hygienists. CFR = Code of Federal Regulations. EPA = Environmental Protection Agency. NIOSH = National Institute of Occupational Safety and Health. OSHA = Occupational Safety and Health Administration. STEL = Short term exposure limit. TLV = Threshold Limit Value. TLV-C = Ceiling Limit Value. TWA = Time Weighted Average.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

SG / EN